

NetAudit Engine — Submission Architecture

SIH26155 · NTRO · AI-Driven Multi-Vendor Network Security Compliance Auditor

Safety boundary: local AI may interpret unfamiliar syntax, but only version-controlled OPA rules produce PASS/FAIL. Invalid or low-confidence extraction becomes **NEEDS_REVIEW**, never compliant.

System architecture

```
Browser → API Gateway → Ingestion → Redis → Parsing Worker
                        ↓           ↓
                MinIO + PostgreSQL SecurityBaseline JSON
                        ↓
                OPA Compliance Engine
                        ↓           ↓
          Remediation + Gate   PDF / JSON / CEF

Optional profiles: reviewed learning/RAG · topology blast radius · Batfish · Ollama
```

Runtime services use an internal Docker network; only the dashboard and gateway publish host ports. Uploads are bounded, MIME checked, credential-redacted, SHA-256 fingerprinted and stored with provenance. RBAC separates admin, operator and auditor access.

Five-stage pipeline

1. **Ingest:** accept configuration text; validate, redact, hash and enqueue.
2. **Normalize:** detect vendor/OS and map CLI to a strict Pydantic SecurityBaseline. Cisco IOS-XE and Fortinet FortiOS are demonstrated.
3. **Learn:** unknown blocks enter an administrator-reviewed mapping queue. Retrieved context never changes a verdict directly.
4. **Evaluate:** OPA applies an 11-control CIS-inspired prototype bundle and retains line-level evidence, severity and versions.
5. **Act and report:** reviewed Jinja2 CLI is preflighted; only SAFE proposals can be approved. Evidence exports as HTML/PDF/JSON/CEF.

Extensibility model

New vendors reuse the normalized schema and service pipeline; they add validated examples/mappings and reviewed remediation templates. New frameworks are versioned OPA bundles over the stable schema, isolating policy updates from vendor syntax.

Prototype Evidence, Limits and Deployment

SIH requirement	Current proof	Next validated milestone
Unified ingestion	Dashboard upload, redaction, SHA-256 and MinIO	Bulk inventory UI and orchestration
AI adaptation	Local structured parsing and reviewed learning APIs	Wire admin learning queue into dashboard
Multi-framework engine	Pluggable, versioned OPA architecture	Authorized CIS/NIST/STIG/ISO content
Actionable reporting	Evidence, severity, remediation, PDF/JSON/CEF	Validate more hardware and firmware
Vendor-agnostic scale	Shared schema; Cisco and Fortinet fixtures	Juniper and Palo Alto validation

Deployment and trust

The default Compose profile is the deterministic presentation core. Optional **advanced** and **model** profiles add learning/topology/Batfish and local Ollama. The system does not push configuration to devices automatically: a human remains responsible for review and change control.

Verification evidence

Gateway 25 passed · Ingestion 20 passed · Compliance 98 passed (6 external integration skips) · Remediation 55 passed · Reporting 14 passed · Schema 8 passed · Scripts 6 passed · frontend type-check and production build passed.

Prototype claim boundary

This prototype demonstrates two vendors, a single-file UI and one 11-control CIS-inspired policy bundle. It does **not** claim CIS certification or complete CIS, NIST, STIG or ISO coverage. The architecture is designed for those additions after content authorization and validation against representative sanitized configurations.

Submission references

Repository: <https://github.com/arc-1103/NetAuditor>
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Generated from the tested submission branch. See README.md and docs/SUBMISSION_CHECKLIST.md for setup and evidence guidance.